

Assignment - 02

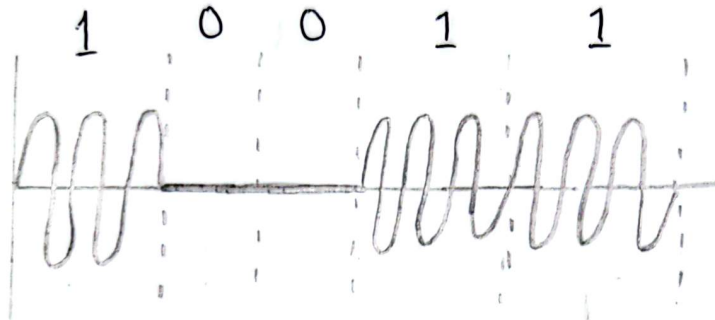
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Section : 13

CSE 320

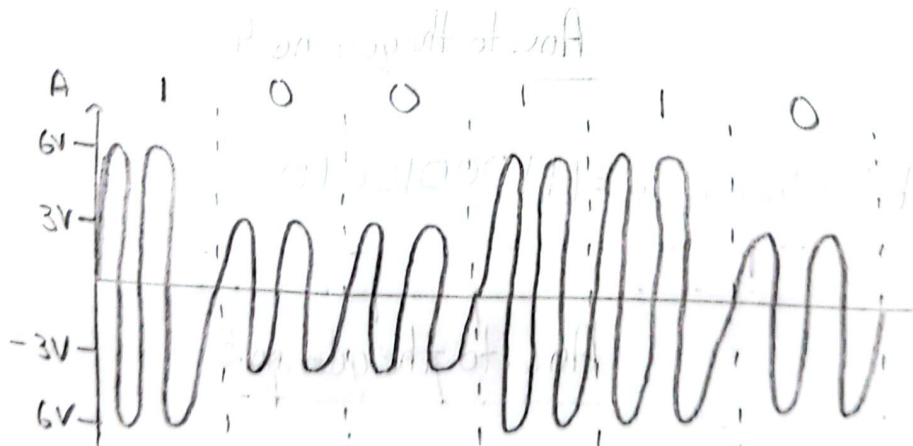
Ans. to the ques. no. 1



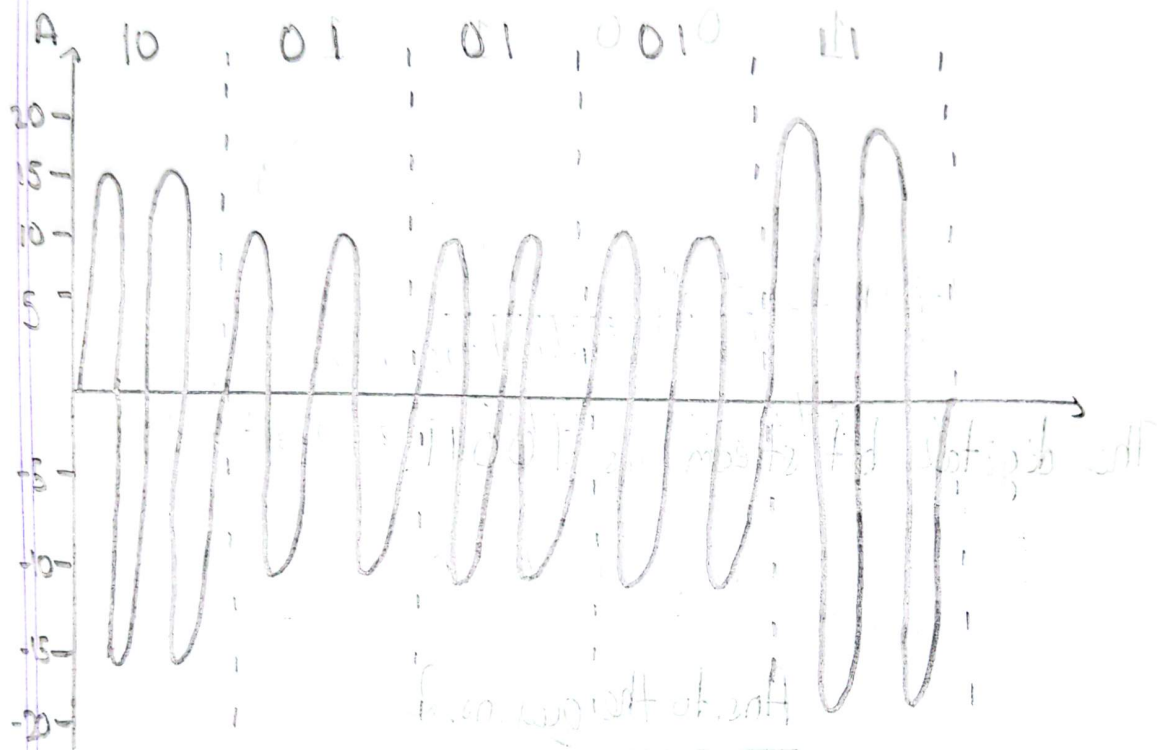
The digital bit stream is 10011.

Ans. to the ques. no. 2

$X = 100110$



Ans. to the ques. no. 3



Ans. to the ques. no. 4

The bit stream is = 11100010010

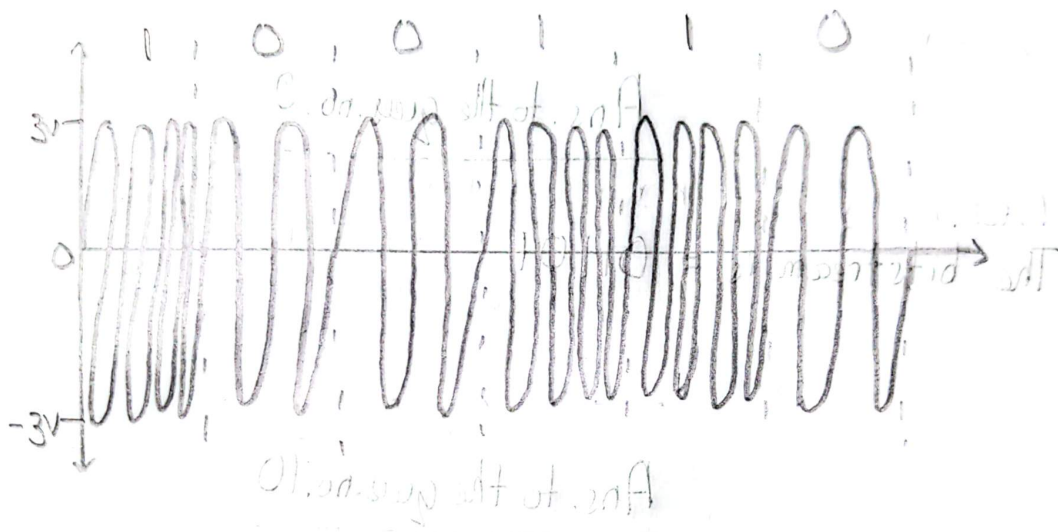
Ans. to the ques. no. 5

The bit stream is = 010011

Ans. to the ques. no. 6

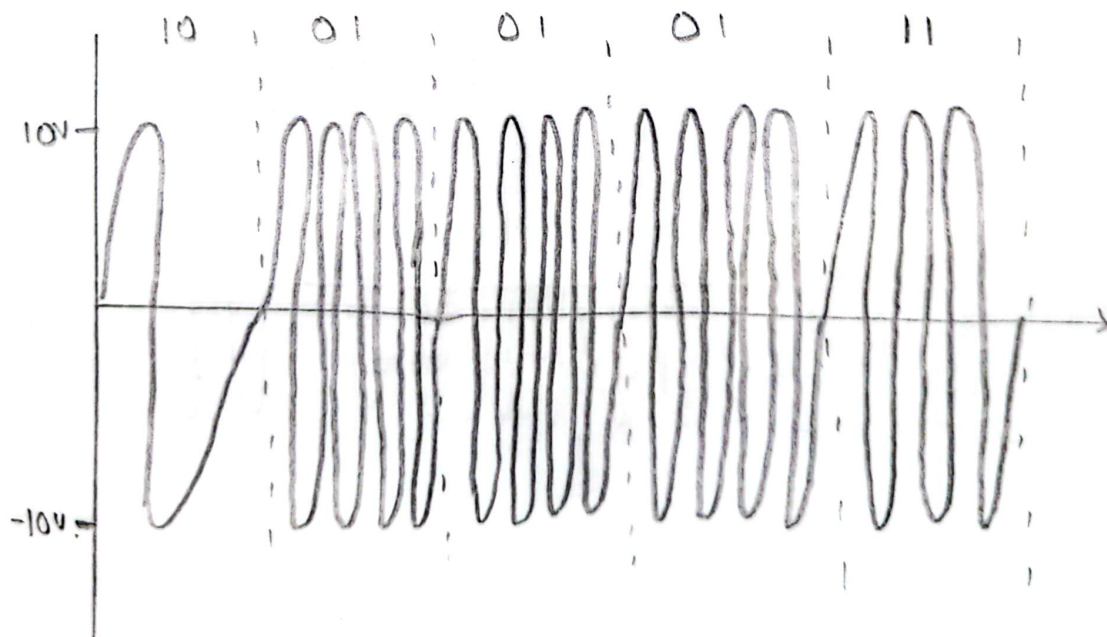
$X = 100110$

$0100111000 = \text{the first zero is isolated at}$



Ans. to the ques. no. 7

$011001 = X$



Ans. to the ques. no. 8

The bitstream is = 0001110010

011001 = 7

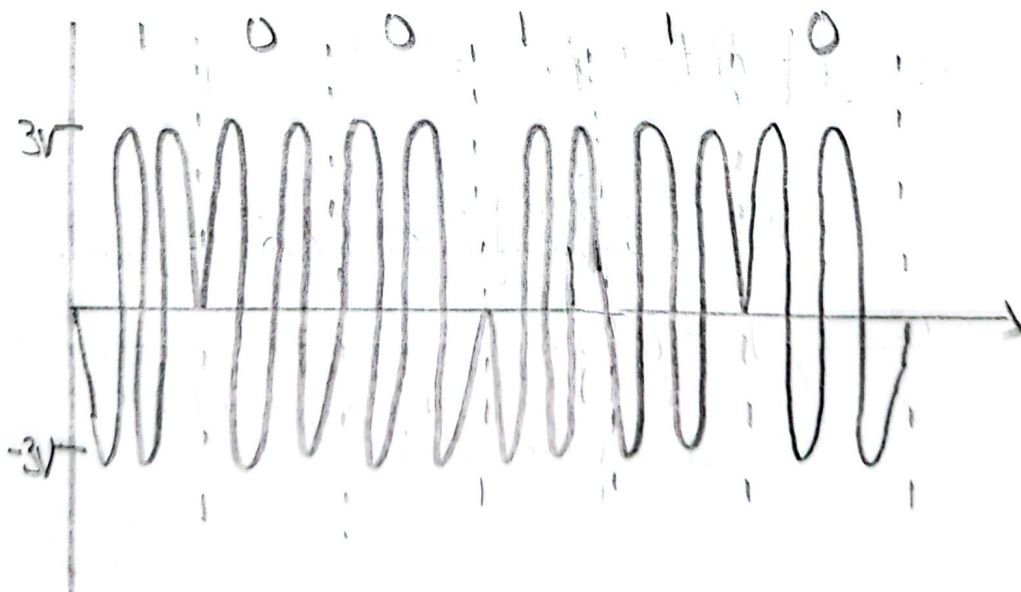
Ans. to the ques. no. 9

The bitstream is = 01101

Ans. to the ques. no. 10

$x = 100110$

For 100110 with 4 bit



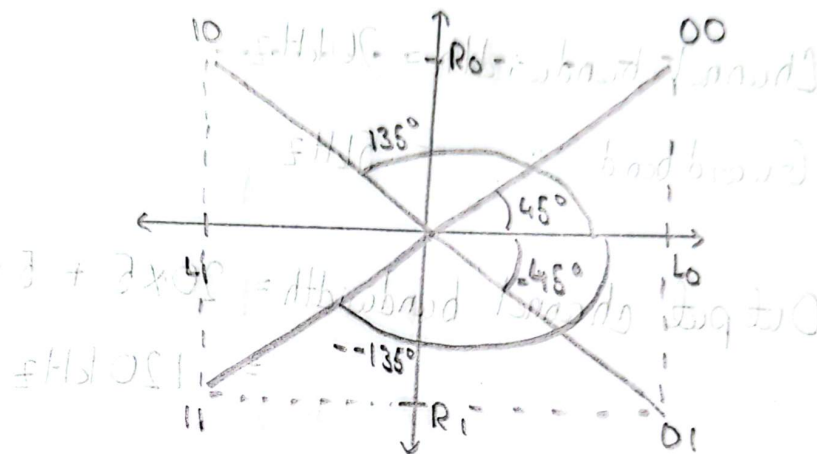
Ans. to the ques. no. 11

$$L_0 = 0^\circ$$

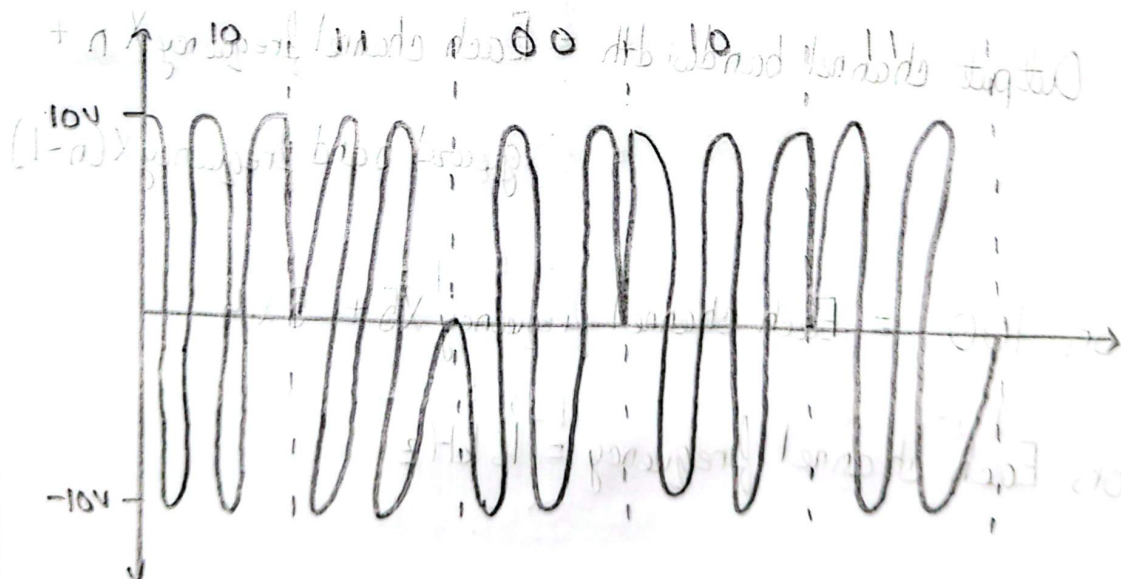
$$L_1 = 180^\circ$$

$$R_0 = 90^\circ$$

$$R_1 = 270^\circ$$



Ans. to the ques. no. 12



$$00 = 180^\circ$$

$$01 = 270^\circ$$

$$10 = 90^\circ$$

$$11 = 0^\circ$$

11. Ans. to the ques. no. 13

$$\text{Channel bandwidth} = 20 \text{ kHz}$$

$$\text{Guard band} = 5 \text{ kHz}$$

$$\begin{aligned}\text{Output channel bandwidth} &= 20 \times 5 + 5 \times (5-1) \\ &= 120 \text{ kHz}\end{aligned}$$

Ans. to the ques. no. 14

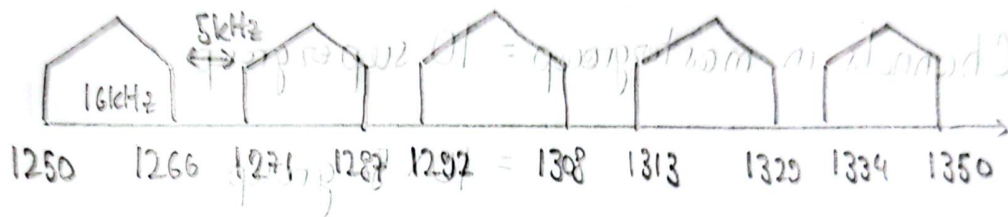
Given

Given

$$\begin{aligned}\text{Output channel bandwidth} &= \text{Each channel frequency} \times n + \\ &\quad \text{guard band frequency} \times (n-1)\end{aligned}$$

$$\text{or, } 100 = \text{Each channel frequency} \times 5 + 5 \times 4$$

$$\text{or, Each channel frequency} = 16 \text{ kHz}$$



The maximum frequency = $1250 + 100$
 $= 1350 \text{ kHz}$

Ans. to the ques. no. 15

Given,

$$100 \text{ kHz} = 20 \times 5 + \text{Guard band} \times 4$$

$$\text{or, Guard band} \times 4 = 0$$

$$\text{or, Guard band} = 0 \text{ kHz}$$

Ans. to the ques. no. 16

Channels in mastergroup = 10 supergroup

$$= 10 \times 5 \text{ group}$$

$$= 10 \times 5 \times 12 \text{ channel}$$

$$= 600 \text{ channels}$$

Here,

$$\begin{aligned} \text{Group Bandwidth} &= 4 \times 12 + 1 \times (12-1) \\ &= 59 \text{ kHz} \end{aligned}$$

$$\text{Supergroup} = 5 \times 59 + 5 \times (5-1)$$

$$= 315 \text{ kHz}$$

$$\text{Mastergroup} = 10 \times 315 + 10 \times (10-1)$$

$$= 3240 \text{ kHz}$$

$\therefore$ Required bandwidth for multiplexing = 3240 kHz